Tensor Factorization of Extension Ideals under Gluing and Huneke–Wiegand Counterexamples with Many Generators

Felipe Santibáñez-Leal
CAOS Research programme, `github.com/fsantibanezleal/CAOS_RESEARCH`

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Abstract

Let $\Gamma = a_1\Gamma_1 + a_2\Gamma_2$ be a gluing of numerical semigroups, $\Bbbk$ a field, and $A = a_1A_1 + a_2A_2$ the extension of relative ideals A_i of Γ_i. With y acting on $\Bbbk[\Gamma_1]$ as t^{a_2} and on $\Bbbk[\Gamma_2]$ as t^{a_1}, we prove that $I_A \otimes_{\Bbbk[\Gamma]} I_{A^*} \cong T_1 \otimes_{\Bbbk[y]} T_2$, where $T_i = I_{A_i} \otimes I_{A_i^*}$. Consequently the torsion lengths satisfy

$$\ell(\tau(I_A \otimes I_{A^*})) = a_1\ell(\tau(T_1)) + a_2\ell(\tau(T_2)) + \sum_{j\geq 1} b_1(j)\, b_2(j),$$

where $b_i(j)$ counts the Jordan blocks of size at least j of y on $\tau(T_i)$, and the numbers of minimal generators multiply. Hence A has the Huneke–Wiegand property if and only if A_1 or A_2 is a non-principal ideal with that property. This re-proves the two gluing propositions of García-Sánchez and Leamer and settles the case they leave open: when both factors are non-principal and the tensor product of each with its dual is torsion-free, so is the tensor product for the extension. Since the Huneke–Wiegand conjecture is now known to be false, this produces counterexample ideals with 2^r minimal generators over symmetric numerical semigroup rings. We give two explicit four-generated examples, over symmetric semigroups with Frobenius numbers 23645 and 53599, built from Pham's counterexample and from an infinite family of counterexamples. Exact computations by two independent routes, the definition of the tensor product and Leamer's partition criterion, confirm the length law on 200 seeded gluings.

1 Introduction

Let R be a one-dimensional Gorenstein local domain. Huneke and Wiegand asked whether a finitely generated torsion-free R-module M with $M \otimes_R \operatorname{Hom}_R(M, R)$ torsion-free must be free [5]. For numerical semigroup rings and monomial ideals, Leamer reduced the question to arithmetic of relative ideals [6], and García-Sánchez and Leamer proved it for monomial ideals of free numerical semigroup rings and for two-generated monomial ideals over complete intersections [3]. Their method rests on gluings $\Gamma = a_1\Gamma_1 + a_2\Gamma_2$: for an extension $a_1A_1 + a_2A_2$ they show that it

inherits the Huneke–Wiegand property from a non-principal factor (their Proposition 9), and that $a_1A_1 + a_2\Gamma_2$ has the property exactly when A_1 does (their Proposition 10). In their final section they ask whether the property for ideals with more generators is invariant under gluing, and whether their Proposition 9 extends to rings of the form $(R_1 \otimes_{\Bbbk} R_2)/(f \otimes 1 - 1 \otimes g)$.

The conjecture is now known to be false in dimension one. Christensen, Gerko and Iyengar gave a non-principal rigid ideal in a one-dimensional Gorenstein domain [1], and Pham gave a two-generated monomial counterexample over a symmetric numerical semigroup of multiplicity 56 and Frobenius number 181 [7]. An infinite family of two-generated counterexamples of multiplicity $24p$, $p \geq 4$, is constructed in [10].

In the presence of counterexamples, the case of an extension whose two factors are both non-principal and both fail the Huneke–Wiegand property is no longer vacuous, and it is not covered by the propositions above. We prove a tensor factorization that decides every extension.

Theorem 1.1. *Let $\Gamma = a_1\Gamma_1 + a_2\Gamma_2$ be a gluing, A_i relative ideals of Γ_i, and $A = a_1A_1 + a_2A_2$. Write $T = I_A \otimes I_{A^*}$ and $T_i = I_{A_i} \otimes I_{A_i^*}$, and let y act on $\Bbbk[\Gamma_1]$ as t^{a_2} and on $\Bbbk[\Gamma_2]$ as t^{a_1}. Then*

1. *$T \cong T_1 \otimes_{\Bbbk[y]} T_2$;*
2. *$\ell(\tau(T)) = a_1\ell(\tau(T_1)) + a_2\ell(\tau(T_2)) + \sum_{j\geq 1} b_1(j)b_2(j)$, where $b_i(j)$ is the number of Jordan blocks of size at least j of y acting on $\tau(T_i)$;*
3. *$\mu(A) = \mu(A_1)\mu(A_2)$;*
4. *A has the Huneke–Wiegand property if and only if A_1 or A_2 is a non-principal ideal with that property.*

Corollary 1.2. *If A_1 and A_2 are two-generated counterexample ideals over symmetric numerical semigroups Γ_1 and Γ_2, and $\Gamma = a_1\Gamma_1 + a_2\Gamma_2$ is a gluing, then $a_1A_1 + a_2A_2$ is a four-generated counterexample ideal over the symmetric semigroup Γ. Iterating, for every $r \geq 1$ there are counterexample ideals with 2^r minimal generators over symmetric numerical semigroup rings.*

Section 2 fixes notation. Section 3 proves part (1), and Section 4 parts (2) to (4). Section 5 derives the propositions of [3], Corollary 1.2, two explicit examples and a partial answer to their invariance question. Section 6 reports exact computations, and Section 7 states the limits of the results.

2 Preliminaries

A numerical semigroup $\Lambda \subseteq \mathbb{N}$ is a submonoid with finite complement; $F(\Lambda)$ is its Frobenius number. A relative ideal is a non-empty set $B \subseteq \mathbb{Z}$, bounded below, with $B + \Lambda \subseteq B$. Its dual is $B^* = \Lambda - B = \{z : z + B \subseteq \Lambda\}$, and B is principal if $B = x + \Lambda$. The fractional ideal I_B of $\Bbbk[\Lambda]$ is spanned by the monomials t^b, $b \in B$, and $I_{B^*} \cong \operatorname{Hom}(I_B, \Bbbk[\Lambda])$ [6, Remark 1.7]. Following [3], a non-principal B has the *Huneke–Wiegand property* if $I_B \otimes I_{B^*}$ has non-zero torsion; by [6, Theorem 1.4] this holds iff the minimal generators of B split into two non-empty sets generating P and Q with $(P + B^*) \cap (Q + B^*) \neq (P \cap Q) + B^*$. A *counterexample ideal* is a non-principal B without the property. We write $\mu(B)$ for the number of minimal generators and $\operatorname{Ap}(B, c) = \{b \in B : b - c \notin B\}$ for $c \in \Lambda$, $c > 0$.

Γ is the *gluing* of Γ_1 and Γ_2 if $\Gamma = a_1\Gamma_1 + a_2\Gamma_2$ with $a_1 \in \Gamma_2$, $a_2 \in \Gamma_1$ and $\gcd(a_1, a_2) = 1$; then $F(\Gamma) = a_1F(\Gamma_1) + a_2F(\Gamma_2) + a_1a_2$, and Γ is symmetric when Γ_1 and Γ_2 are [2, 8]. The *extension* of $A_i \subseteq \Gamma_i$ is $A = a_1A_1 + a_2A_2$; it is a relative ideal of Γ, generated by $a_1x + a_2z$ over

generators x of A_1 and z of A_2, and $A^* = a_1A_1^* + a_2A_2^*$ [3, Lemma 8]. Throughout, $R_i = \Bbbk[\Gamma_i]$, $R = \Bbbk[\Gamma]$, $S = \Bbbk[y]$ with $y \mapsto t^{a_2}$ in R_1 and $y \mapsto t^{a_1}$ in R_2. All rings and modules are graded by the exponent; for graded modules of finite length, $\ell = \dim_{\Bbbk}$.

3 Extensions are tensor products

Lemma 3.1. *Let B be a relative ideal of Λ and $c \in \Lambda$, $c > 0$. Then I_B is a free $\Bbbk[t^c]$-module with basis $\{t^b : b \in \mathrm{Ap}(B, c)\}$, and $\mathrm{Ap}(B, c)$ has exactly one element in each residue class modulo c.*

Proof. Each residue class modulo c meets B in a non-empty set that is bounded below and closed under adding c. With b_r its least element, the class is $b_r + c\mathbb{N}$, so $I_B = \bigoplus_r t^{b_r}\Bbbk[t^c]$. □

Proposition 3.2. *The assignment $t^x \otimes t^z \mapsto t^{a_1x+a_2z}$ defines an isomorphism of $R_1 \otimes_S R_2$-modules $I_{A_1} \otimes_S I_{A_2} \to I_A$. For $A_i = \Gamma_i$ it is a ring isomorphism $R_1 \otimes_S R_2 \cong R$, and $I_{A^*} \cong I_{A_1^*} \otimes_S I_{A_2^*}$.*

Proof. The map is S-balanced, since $(t^{a_2}t^x) \otimes t^z$ and $t^x \otimes (t^{a_1}t^z)$ both go to $t^{a_1x+a_1a_2+a_2z}$, and it is compatible with both actions. By Lemma 3.1 the source is S-free on $t^x \otimes t^z$ with $x \in \mathrm{Ap}(A_1, a_2)$ and $z \in \mathrm{Ap}(A_2, a_1)$, so it has the $\Bbbk$-basis $y^n(t^x \otimes t^z)$, sent to $t^{a_1x+a_2z+na_1a_2}$. Every element of A is $a_1x' + a_2z'$ with $x' = x + ia_2$ and $z' = z + ja_1$ for Apéry elements x, z and $i, j \geq 0$, hence equals $a_1x + a_2z + (i + j)a_1a_2$; so the map is onto. If $a_1x + a_2z + na_1a_2 = a_1x' + a_2z' + n'a_1a_2$, reduction modulo a_1 gives $z \equiv z'$, hence $z = z'$ by Lemma 3.1; likewise $x = x'$, and then $n = n'$. Distinct basis vectors therefore go to distinct monomials. The last statement follows from $A^* = a_1A_1^* + a_2A_2^*$. □

Proposition 3.3. *$T \cong T_1 \otimes_S T_2$ as modules over $R \cong R_1 \otimes_S R_2$.*

Proof. For commutative S-algebras R_i and R_i-modules M_i, N_i the assignment $(m_1 \otimes m_2) \otimes (n_1 \otimes n_2) \mapsto (m_1 \otimes n_1) \otimes (m_2 \otimes n_2)$ gives an isomorphism $(M_1 \otimes_S M_2) \otimes_{R_1 \otimes_S R_2} (N_1 \otimes_S N_2) \cong (M_1 \otimes_{R_1} N_1) \otimes_S (M_2 \otimes_{R_2} N_2)$: both sides are the quotient of $M_1 \otimes_S M_2 \otimes_S N_1 \otimes_S N_2$ by the relations $r_im_i \otimes n_i = m_i \otimes r_in_i$, and the inverse assignment is well defined on them. Apply it with $M_i = I_{A_i}$, $N_i = I_{A_i^*}$ and Proposition 3.2. □

4 The torsion-length law

Lemma 4.1. *S embeds in R by $y \mapsto t^{a_1a_2}$, R is a finite S-module, and R_1 and R_2 are free over S of ranks a_2 and a_1. For a finitely generated module over R, R_1 or R_2, the torsion submodule equals the S-torsion submodule.*

Proof. The ranks come from Lemma 3.1. A non-zero element of S stays non-zero in each domain, so S-torsion is torsion. Conversely, if $rm = 0$ with $r \neq 0$, take a monic relation $r^d + s_{d-1}r^{d-1} + \cdots + s_0 = 0$ over S of least degree; then $s_0 \neq 0$ and $s_0m = -(r^{d-1} + \cdots + s_1)rm = 0$. □

Proof of Theorem 1.1. Part (1) is Proposition 3.3. For (2), each T_i is finitely generated over S and has rank one over R_i, so over the graded principal ideal domain S we have $T_1 \cong S^{a_2} \oplus \tau_1$ and $T_2 \cong S^{a_1} \oplus \tau_2$ with $\tau_i = \tau(T_i)$ by Lemma 4.1. Then

$$T \cong S^{a_1a_2} \oplus \tau_1^{a_1} \oplus \tau_2^{a_2} \oplus (\tau_1 \otimes_S \tau_2).$$

The last three summands are S-torsion and the first is free, so their sum is $\tau(T)$. Each τ_i has finite length and y raises degree, so y is nilpotent on it: $\tau_1 \cong \bigoplus S/(y^{e_i})$ and $\tau_2 \cong \bigoplus S/(y^{f_j})$ up to shifts. Since $S/(y^e) \otimes_S S/(y^f) \cong S/(y^{\min(e,f)})$, the length of $\tau_1 \otimes_S \tau_2$ is $\sum_{i,j} \min(e_i, f_j) = \sum_{j \geq 1} b_1(j)b_2(j)$.

For (3), let $\mathfrak{m}$ be the graded maximal ideal. Then $R/\mathfrak{m} = \Bbbk = (R_1/\mathfrak{m}_1) \otimes_S (R_2/\mathfrak{m}_2)$ with y acting by zero, and the isomorphism of Proposition 3.3 applied to $N_i = \Bbbk$ gives $I_A \otimes_R \Bbbk \cong (I_{A_1} \otimes_{R_1} \Bbbk) \otimes_\Bbbk (I_{A_2} \otimes_{R_2} \Bbbk)$, of dimension $\mu(A_1)\mu(A_2)$.

For (4), a principal factor has $T_i \cong R_i$, which is torsion-free, and a non-principal factor has $\tau(T_i) \neq 0$ exactly when it has the Huneke–Wiegand property. Since $a_1, a_2 \geq 1$, part (2) gives $\tau(T) \neq 0$ iff $\tau(T_1) \neq 0$ or $\tau(T_2) \neq 0$, and part (3) gives that A is non-principal iff a factor is. □

5 Consequences

Corollary 5.1 ([3, Propositions 9 and 10])**.** *If A_1 is a non-principal ideal with the Huneke–Wiegand property, every extension $a_1A_1 + a_2B$ has it. The extension $a_1A_1 + a_2\Gamma_2$ has it if and only if A_1 has.*

Corollary 5.2. *If A_1 and A_2 are both counterexample ideals, then $a_1A_1+a_2A_2$ is a counterexample ideal with $\mu(A_1)\mu(A_2)$ minimal generators.*

Corollary 1.2 follows from Corollary 5.2 and the symmetry of gluings of symmetric semigroups. For the simple gluings $m\mathbb{N} + q\Gamma$ with $\gcd(q, m) = 1$ and m the multiplicity of Γ, the case $\Gamma_2 = \mathbb{N}$, $A_2 = \mathbb{N}$ of Theorem 1.1, already given by [3, Proposition 10], turns the family of [10] into a two-parameter family of symmetric counterexamples of multiplicity $24p$ and embedding dimension $11p$, for every $p \geq 4$ and every $q \geq 1$ coprime to $24p$.

Example 5.3. Let Γ_P be Pham's semigroup, minimally generated by 56, 57, 58, 63, 64, the integers from 70 to 83, and 87, 89, 90, 93, 95, 96, 97 (multiplicity 56, Frobenius number 181), with the counterexample ideal $(0, 14)$ [7]. For the gluing $\Gamma = 56\Gamma_P + 57\Gamma_P$, the extension $56(0, 14) + 57(0, 14)$ is the relative ideal with minimal generators $0, 784, 798, 1582$. Γ is symmetric with Frobenius number $56 \cdot 181 + 57 \cdot 181 + 56 \cdot 57 = 23645$, and the extension is a four-generated counterexample.

Example 5.4. Let C_4 be the member $p = 4$ of the family of [10], of multiplicity 96 and Frobenius number 311, with counterexample ideal $(0, 24)$. For $\Gamma = 96\Gamma_P + 89C_4$, the extension $96(0, 14) + 89(0, 24)$ has minimal generators $0, 1344, 2136, 3480$, and Γ is symmetric with Frobenius number $96 \cdot 181 + 89 \cdot 311 + 96 \cdot 89 = 53599$.

Corollary 5.5. *If every non-principal relative ideal of Γ_1 and of Γ_2 has the Huneke–Wiegand property, then so does every non-principal extension ideal of $a_1\Gamma_1 + a_2\Gamma_2$.*

This answers the invariance question of [3, Section 4] for extension ideals. Relative ideals of Γ that are not extensions are not covered.

Remark 5.6. For monomial ideals, the module $(I_{A_1} \otimes_\Bbbk I_{A_2})/(t^{a_2} \otimes 1 - 1 \otimes t^{a_1})$ of the ring-level question in [3, Section 4] is $I_{A_1} \otimes_S I_{A_2} = I_A$, so Theorem 1.1 answers that question for monomial ideals. A statement for arbitrary modules would need the dual of $M_1 \otimes_S M_2$ to factor over S, which is not established here.

6 Computational checks

The statements above are proved; the computations below test them and their transcription. All arithmetic is exact.

Two routes. The torsion of $I_B \otimes I_{B^*}$ is computed from the definition: in degree d the module is spanned by $e_x \otimes e_z$ with $x \in B$, $z \in B^*$, $x + z = d$, modulo $e_{x+g} \otimes e_z = e_x \otimes e_{z+g}$ for generators g, so its dimension is the number of connected components of a finite graph and the torsion in degree d has one dimension fewer. Every pair joins a pair $(x_i, d - x_i)$ with x_i a generator of B, and two such pairs join through $w = \max_i x_i + F(\Lambda) + 1$ once $d - w$ exceeds the conductor of B^*, which bounds the degrees to scan. Independently, the Huneke–Wiegand property is decided by the partition criterion of [6].

Seeded family. For 200 gluings with random factors of Frobenius number at most 10, gluing parameters at most 7 and ideals with at most three generators, the directly computed torsion length of the extension equals part (2) of Theorem 1.1 computed from the factors alone, including the Jordan types of y; in 33 of the cases both factors have torsion, so the cross term is non-zero. The partition criterion agrees with part (4), and part (3) and the Hilbert series identity $H_{I_A}(z) = H_{I_{A_1}}(z^{a_1}) H_{I_{A_2}}(z^{a_2})(1 - z^{a_1 a_2})$ hold in every case.

Examples and controls. The partition criterion confirms Examples 5.3 and 5.4, and that $(0, 14)$ and $(0, 24)$ are counterexample ideals of Γ_P and C_4. Replacing the second factor of Example 5.3 by the ideal $(0, 1)$ of Γ_P, which has the property, yields an extension with the property. For 30 random pairs with $a_1 \notin \Gamma_2$ the Hilbert series identity fails, as it must without the gluing hypothesis. A second implementation sharing no code with the first recomputes Example 5.3 and the torsion lengths of 25 of the 200 cases, with full agreement. The code and data are in the repository named on the first page.

7 Scope

The results concern monomial (relative) ideals of numerical semigroup rings over any field, graded or completed. Theorem 1.1 decides extension ideals of a gluing only; whether the gluing of two semigroups whose non-principal ideals all have the Huneke–Wiegand property can carry a counterexample ideal that is not an extension remains open. The sources consulted for the factorization and the length law were [3, 4, 6, 9]; none of them states these results, which is not a proof that they are new.

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